

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2310

Southbase 138 kV
July 20, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2310. The Interconnection Customer proposes a 137.8 MW solar photovoltaic facility interconnecting at the Southbase 138 kV substation in the Pennsylvania area. The request was received on April 29, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 606 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$19,049,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
PATTERSON / FREEPORT 138kV	MOUNTAIN 345/138 kV TX	110.0	310	6.1
GREENVILLE STEAM / CLAYTON 138kV Line	KINGS / CLAYTON 138kV	110.0	169	9.5
PATTERSON / GREINWICH 138kV	MOUNTAIN 345-138 kV Auto	106.8	269	6.1
KINGS / CLATON 138kV Line	GREENVILLE STEAM / KINGS 138kV Line	104.3	388	32.1
RAINEY / MOUNTAIN 138kV	LITTLE CHUTE / TALENENERGY MARTINS CREEK LLC HARRISBURG 345kV	103.4	445	5.7
GREENVILLE STEAM / KINGS 138kV	CLAYTON / KINGS 138kV line	102.3	334	28.5
BEMIDJI / METALCAN 138kV L	LITTLE CHUTE / TALENENERGY MARTINS CREEK LLC HARRISBURG 345kV	100.8	406	7.9
CARDWELL / BEMIDJI 138kV L	LITTLE CHUTE / TALENENERGY MARTINS CRK LLC HARRISBURG 345kV	100.6	422	7.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PATTERSON / FREEPORT 138kV	Rebuild line with higher-capacity conductor	\$4,115,000	26% (\$1,085,000)
GREENVILLE STEAM / CLAYTON 138kV Line	Reconductor line	\$5,099,000	53% (\$2,720,000)
PATTERSON / GREINWICH 138kV	Reconductor line	\$1,243,000	37% (\$463,000)
KINGS / CLATON 138kV Line	Reconductor line	\$4,034,000	82% (\$3,305,000)
RAINEY / MOUNTAIN 138kV	Reconductor line	\$2,376,000	41% (\$968,000)
GREENVILLE STEAM / KINGS 138kV	Reconductor line	\$4,449,000	86% (\$3,824,000)
BEMIDJI / METALCAN 138kV L	Reconductor line	\$668,000	54% (\$363,000)
CARDWELL / BEMIDJI 138kV L	Reconductor line	\$11,616,000	54% (\$6,321,000)
	Total allocated to project		\$19,049,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.